

BG770A-GL&BG95xA-GL

MQTT Application Note

LPWA Module Series

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About the Document

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1 Introduction

Message Queuing Telemetry Transport (MQTT) is a broker-based publish/subscribe messaging protocol designed to be open, simple, lightweight, and easy to implement. It is designed for connections with remote locations where a "small code footprint" is required, or the network bandwidth is limited.

This document explains how to use the MQTT function of Quectel BG770A-GL, BG950A-GL and BG951A-GL modules through AT commands.

2 MQTT Data Interaction

This chapter illustrates the data interaction mechanism of the MQTT function.

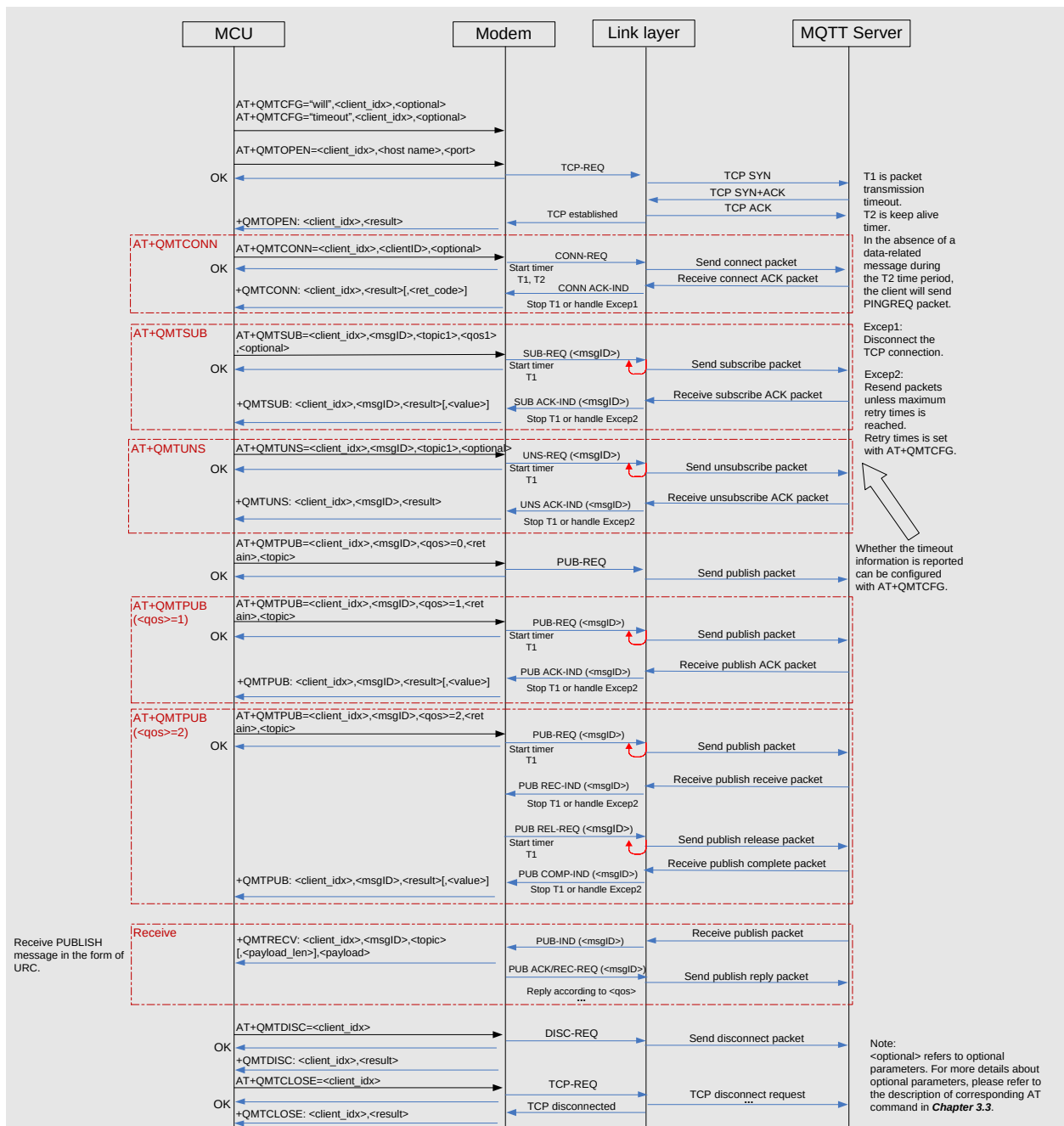


Figure 1: MQTT Data Interaction Diagram

3 MQTT AT Commands

The AT commands for using the MQTT function are presented in this chapter.

3.1. AT Command Introduction

3.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

3.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 1: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of the corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of the corresponding command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.

Execution Command **AT+<cmd>**

Return a specific information parameter or perform a specific action.

3.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

3.3. Description of MQTT AT Commands

3.3.1. AT+QMTCFG Configure Optional Parameters of MQTT

This command configures optional parameters of MQTT.

AT+QMTCFG Configure Optional Parameters of MQTT	
Test Command AT+QMTCFG=?	Response +QMTCFG: "version",(range of supported <client_idx>s),(list of supported <vsn>s) +QMTCFG: "pdpclid",(range of supported <client_idx>s),(range of supported <cid>s) +QMTCFG: "ssl",(range of supported <client_idx>s),(list of supported <SSL_enable>s),(range of supported <ctx_index>s) +QMTCFG: "keepalive",(range of supported <client_idx>s),(range of supported <keep_alive_time>s) +QMTCFG: "session",(range of supported <client_idx>s),(list of supported <clean_session>s) +QMTCFG: "timeout",(range of supported <client_idx>s),(range of supported <pkt_timeout>s),(range of supported <retry_times>s),(list of supported <timeout_notice>s) +QMTCFG: "will",(range of supported <client_idx>s),(list of supported <will_fg>s),(range of supported <will_qos>s),(list of supported <will_retain>s),<will_topic>,<will_message> +QMTCFG: "recv/mode",(range of supported <client_idx>s),(list of supported <msg_rcv_mode>s),(list of supported <msg_len_enable>s) +QMTCFG: "aliauth",(range of supported <client_idx>s),<product_key>,<device_name>,<device_secret>

	OK
Write Command Query/set the MQTT protocol version AT+QMTCFG="version",<client_idx>,<vsn>]	Response If the optional parameter is omitted, query the MQTT protocol version: +QMTCFG: "version",<vsn> OK If the optional parameter is specified, set the MQTT protocol version: OK If there is any error: ERROR
Write Command Query/set the PDP to be used by the MQTT client AT+QMTCFG="pdpcid",<client_idx>,<cid>]	Response If the optional parameter is omitted, query the PDP used by the MQTT client: +QMTCFG: "pdpcid",<cid> OK If the optional parameter is specified, set the PDP to be used by the MQTT client: OK If there is any error: ERROR
Write Command Query/set the MQTT SSL mode and SSL context index AT+QMTCFG="ssl",<client_idx>,<SSL_enable>,<ctx_index>]]	Response If the optional parameters are omitted, query the MQTT SSL mode and SSL context index: +QMTCFG: "ssl",<SSL_enable>,<ctx_index>] OK If any of the optional parameters is specified, set the MQTT SSL mode and SSL context index: OK If there is any error: ERROR
Write Command Query/set the keep-alive time AT+QMTCFG="keepalive",<client_idx>,<keep_alive_time>]	Response If the optional parameter is omitted, query the keep-alive time: +QMTCFG: "keepalive",<keep_alive_time> OK

	If the optional parameter is specified, set the keep-alive time: OK If there is any error: ERROR
Write Command Query/set the session type AT+QMTCFG="session",<client_idx>[,<clean_session>]	Response If the optional parameter is omitted, query the session type: +QMTCFG: "session",<clean_session> OK If the optional parameter is specified, set the session type: OK If there is any error: ERROR
Write Command Query/set the message delivery timeout AT+QMTCFG="timeout",<client_idx>[,<pkt_timeout>[,<retry_times>][,<timeout_notice>]]	Response If the optional parameters are omitted, query the message delivery timeout: +QMTCFG: "timeout",<pkt_timeout>,<retry_times>,<timeout_notice> OK If any of the optional parameters is specified, set the message delivery timeout: OK If there is any error: ERROR
Write Command Query/set the Will information AT+QMTCFG="will",<client_idx>[,<will_fg>[,<will_qos>,<will_retain>,<will_topic>,<will_message>]]	Response If the optional parameters are omitted, query the Will information: +QMTCFG: "will",<will_fg>[,<will_qos>,<will_retain>,<will_topic>,<will_message>] OK If any of the optional parameters is specified, set the Will information: OK If there is any error: ERROR
Write Command Query/set the MQTT	Response If the optional parameters are omitted, query the MQTT message

message receiving mode when the data are received from server AT+QMTCFG="recv/mode", <client_idx>[,<msg_rcv_m ode>[,<msg_len_enable>]]	receiving mode. +QMTCFG: "recv/mode",<msg_rcv_mode>[,<msg_len_enable>] OK If any of the optional parameters is specified, set the MQTT message receiving mode: OK If there is any error: ERROR
Write Command Query/set Alibaba device information for AliCloud AT+QMTCFG="aliauth",<cli ent_idx>[,<product_key>,< device_name>,<device_sec ret>]	Response If the optional parameters are omitted, query the device information: [+QMTCFG: "aliauth",<product_key>,<device_name>,<device_sec ret>] OK If the optional parameters are specified, set the device information: OK If there is any error: ERROR
Maximum Response Time	Determined by the network.
Characteristics	The command takes effect immediately. The configurations will not be saved.

Parameter

<client_idx>	Integer type. MQTT client identifier. Range: 0–5.
<vsn>	Integer type. MQTT protocol version. 3 MQTT v3.1 4 MQTT v3.1.1
<cid>	Integer type. PDP to be used by the MQTT client. Range: 1–5. Default value: 1
<will_fg>	Integer type. Whether to configure the Will flag. 0 Ignore the Will flag configuration 1 Require the Will flag configuration
<will_qos>	Integer type. QoS level of message delivery. 0 At most once 1 At least once 2 Exactly once
<will_retain>	Integer type. Will Retain flag is only used for PUBLISH messages.

	0 When a client sends a PUBLISH message to a server, the server will not retain the message after it has been delivered to the current subscribers. 1 When a client sends a PUBLISH message to a server, the server should retain the message after it has been delivered to the current subscribers.
<will_topic>	String type. Will topic. Maximum length: 255 bytes.
<will_message>	String type. The Will message defines the content of the message published on the Will topic if the client is unexpectedly disconnected. It can be a zero-length message. Maximum length: 255 bytes.
<pkt_timeout>	Integer type. Packet delivery timeout. Range: 1–60. Default value: 5. Unit: second.
<retry_times>	Integer type. Retry times when packet delivery times out. Range: 0–10. Default value: 3.
<timeout_notice>	Integer type. Whether to report timeout message when transmitting a packet.
	0 Do not report 1 Report
<clean_session>	Integer type. Configures the session type.
	0 The server must store the subscriptions of the client after it disconnects. 1 The server must discard any previously maintained information about the client after it disconnects and treat the connection as “clean”.
<keep_alive_time>	Integer type. Keep-alive time. Range: 0–3600. Default value: 120. Unit: second. It defines the maximum interval between messages received from a client. If the server does not receive a message from the client within 1.5 times of the keep-alive time value, it disconnects the client as if the client sent a DISCONNECT message. If the keep-alive time is 0, this means that the server is not required to disconnect the client on the grounds of inactivity.
<SSL_enable>	Integer type. MQTT SSL mode.
	0 Use normal TCP connection for MQTT 1 Use SSL TCP secure connection for MQTT
<ctx_index>	Integer type. SSL context index. Range: 0–5.
<msg_rcv_mode>	Integer type. MQTT message receiving mode.
	0 MQTT message received from server will be contained in URC 1 MQTT message received from server will not be contained in URC
<msg_len_enable>	Integer type.
	0 Length of MQTT message received from server will not be contained in URC 1 Length of MQTT message received from server will be contained in URC
<product_key>	Product key issued by AliCloud.
<device_name>	Device name issued by AliCloud.
<device_secret>	Device secret key issued by AliCloud.

NOTE

- If <will_fg>=1, then <will_qos>, <will_retain>, <will_topic> and <will_message> must be specified. Otherwise, they are omitted.
- <clean_session>=0 is only effective when the server supports the operation.

3. If MQTT connection is configured to SSL mode, **<ctx_index>** must be specified, and **AT+QSSLCFG** must be used to configure the SSL version, cipher suite, secure level, CA certificate, client certificate, client key and ignorance of RTC time, which will be used in MQTT SSL handshake procedure. For more information on **AT+QSSLCFG**, see **document [3]**.
4. Care must be taken to ensure that message delivery does not time out while the sending process is in progress.
5. **AT+QMTCFG="aliauth"** is only used for AliCloud. If it is configured, **<username>** and **<password>** can be omitted in **AT+QMTCONN**. Once set, a specified session can only be used for AliCloud until reboot.

3.3.2. AT+QMTOPEN Open a Network Connection for MQTT Client

This command opens a network connection for an MQTT client.

AT+QMTOPEN Open a Network Connection for MQTT Client

Test Command AT+QMTOPEN=?	Response +QMTOPEN: (range of supported <client_idx>s), <host_name> , (range of supported <port>s) OK
Read Command AT+QMTOPEN?	Response [+QMTOPEN: <client_idx>,<host_name>,<port>] OK
Write Command AT+QMTOPEN=<client_idx>,<host_name>,<port>	Response OK +QMTOPEN: <client_idx>,<result> If there is any error: ERROR
Maximum Response Time	Determined by the network
Characteristics	/

Parameter

<client_idx>	Integer type. MQTT client identifier. Range: 0–5.
<host_name>	String type. Server address. It could be an IP address or a domain name. Maximum size: 128 bytes.
<port>	Integer type. Server port number. Range: 0–65535.
<result>	Integer type. Command execution result. -1 Failed to open a network

- | | |
|---|------------------------------------|
| 0 | Network opened successfully |
| 1 | Wrong parameter |
| 2 | MQTT client identifier is occupied |
| 3 | Failed to activate PDP |
| 4 | Failed to parse domain name |
| 5 | Network connection error |

3.3.3. AT+QMTCLOSE Close a Network Connection for MQTT Client

This command closes a network connection for an MQTT client.

AT+QMTCLOSE Close a Network Connection for MQTT Client

Test Command AT+QMTCLOSE=?	Response +QMTCLOSE: (range of supported <client_idx>s) OK
Write Command AT+QMTCLOSE=<client_idx>	Response OK +QMTCLOSE: <client_idx>,<result> If there is any error: ERROR
Maximum Response Time	Determined by network
Characteristics	/

Parameter

<client_idx>	Integer type. MQTT client identifier. Range: 0–5.
<result>	Integer type. Command execution result.
-1	Failed to close the network
0	Network closed successfully

3.3.4. AT+QMTCONN Connect a Client to MQTT Server

This command is used when a client requests a connection to the MQTT server. When a TCP/IP socket connection is established between a client and a server, a protocol level session must be created using a CONNECT flow.

AT+QMTCONN Connect a Client to MQTT Server	
Test Command AT+QMTCONN=?	Response +QMTCONN: (range of supported <client_idx>s),<clientID>,<username>,<password> OK
Read Command AT+QMTCONN?	Response [+QMTCONN: <client_idx>,<state>] OK
Write Command AT+QMTCONN=<client_idx>,<clientID>[,<username>[,<password>]]	Response OK +QMTCONN: <client_idx>,<result>[,<ret_code>] If there is any error: ERROR
Maximum Response Time	<pkt_timeout> (default 5 s), determined by the network
Characteristics	/

Parameter

<client_idx>	Integer type. MQTT client identifier. Range: 0–5.
<clientID>	Client identifier string.
<username>	String type. Client username. It can be used for authentication.
<password>	String type. The password corresponding to the client username. It can be used for authentication.
<result>	Integer type. Command execution result. 0 Packet sent successfully and ACK received from the server 1 Packet retransmission 2 Failed to send the packet
<state>	Integer type. MQTT connection state. 1 MQTT is initializing 2 MQTT is connecting 3 MQTT is connected 4 MQTT is disconnecting
<ret_code>	Integer type. Connection status return code. 0 Connection Accepted 1 Connection Refused: Unacceptable Protocol Version 2 Connection Refused: Identifier Rejected 3 Connection Refused: Server Unavailable

	4	Connection Refused: Bad Username or Password
	5	Connection Refused: Not Authorized
<pkt_timeout>	Integer type. Packet delivery timeout. Range: 1–60. Default value: 5. Unit: second.	

NOTE

1. If a client with the same Client ID is already connected to the server, the "older" client must be disconnected by the server before completing the CONNECT flow of the new client.
2. **AT+QMTCFG="aliauth"** is only used for AliCloud. If it is configured, then **<username>** and **<password>** in **AT+QMTCONN** can be omitted.

3.3.5. AT+QMTDISC Disconnect a Client from MQTT Server

This command requests a disconnection from MQTT server by a client. A **DISCONNECT** message is sent from the client to the server to indicate that it is about to close its TCP/IP connection.

AT+QMTDISC Disconnect a Client from MQTT Server

Test Command AT+QMTDISC=?	Response +QMTDISC: (range of supported <client_idx> s) OK
Write Command AT+QMTDISC=<client_idx>	Response OK +QMTDISC: <client_idx> , <result> If there is any error: ERROR
Maximum Response Time	Determined by the network
Characteristics	/

Parameter

<client_idx>	Integer type. MQTT client identifier. Range: 0–5.
<result>	Integer type. Command execution result.
	-1 Failed to close the connection
	0 Connection closed successfully

NOTE

Disconnecting a client closes the socket.

3.3.6. AT+QMTSUB Subscribe to Topics

This command subscribes to one or more topics. A **SUBSCRIBE** message is sent to the server by a client to register interest in one or more topic names. Messages published on these topics are delivered from the server to the client as **PUBLISH** messages.

AT+QMTSUB Subscribe to Topics

Test Command AT+QMTSUB=?	Response +QMTSUB: (range of supported <client_idx>s),(range of supported <msgid>s), <topic> ,(range of supported <qos>s) OK
Write Command AT+QMTSUB=<client_idx>,<msgid>,<topic1>,<qos1>[,<topic2>,<qos2>...]	Response OK +QMTSUB: <client_idx>,<msgid>,<result>[,<value>] If there is any error: ERROR
Maximum Response Time	<pkt_timeout> × <retry_times> (default 15 s), determined by network
Characteristics	The command takes effect immediately. The configurations will not be saved.

Parameter

<client_idx>	Integer type. MQTT client identifier. Range: 0–5.
<msgid>	Integer type. Message identifier of a packet. Range: 1–65535.
<topic>	String type. Topic that the client wants to subscribe to or unsubscribe from.
<qos>	Integer type. QoS level at which the client wants to publish messages. 0 At most once 1 At least once 2 Exactly once
<result>	Integer type. Command execution result. 0 Packet sent successfully and ACK received from the server 1 Packet retransmission 2 Failed to send a packet
<value>	If <result> is 0, it is a vector of granted QoS levels.

	If <result> is 1, it is the number of times the packet has been retransmitted. If <result> is 2, it will not be presented.
<pkt_timeout>	Integer type. Packet delivery timeout. Range: 1–60. Default value: 5. Unit: second.
<retry_times>	Integer type. Retry times when packet delivery times out. Range: 0–10. Default value: 3.

NOTE

The **<msgID>** is only present in messages where the QoS bits in the fixed header indicate QoS levels 1 or 2. It must be unique among the set of "inflight" messages in a particular communication direction. It typically increases by 1 from one message to the next.

3.3.7. AT+QMTUNS Unsubscribe from Topics

This command unsubscribes from one or more topics. An **UNSUBSCRIBE** message is sent by the client to the server to unsubscribe from the named topics.

AT+QMTUNS Unsubscribe from Topics	
Test Command AT+QMTUNS=?	Response +QMTUNS: (range of supported <client_idx> s),(range of supported <msgID> s), <topic> OK
Write Command AT+QMTUNS=<client_idx>,<msgID>,<topic1>[,<topic2>...]	Response OK +QMTUNS: <client_idx>,<msgID>,<result> If there is any error: ERROR
Maximum Response Time	<pkt_timeout> × <retry_times> (default 15 s), determined by network
Characteristics	The command takes effect immediately. The configurations will not be saved.

Parameter

<client_idx>	Integer type. MQTT client identifier. Range: 0–5.
<msgID>	Integer type. Message identifier of a packet. Range: 1–65535.
<topic>	String type. Topic that the client wants to subscribe to or unsubscribe from.
<result>	Integer type. Command execution result. 0 Packet sent successfully and ACK received from the server

	1 Packet retransmission
	2 Failed to send a packet
<pkt_timeout>	Integer type. Packet delivery timeout. Range: 1–60. Default value: 5. Unit: second.
<retry_times>	Integer type. Retry times when packet delivery times out. Range: 0–10. Default value: 3.

3.3.8. AT+QMTPUB Publish Messages (Data Mode)

This command publishes messages by a client to a server for distribution to interested subscribers. Each **PUBLISH** message is associated with a topic name. If a client subscribes to one or more topics, any message published on those topics is sent by the server to the client as a **PUBLISH** message.

AT+QMTPUB Publish Messages (Data Mode)	
Test Command AT+QMTPUB=?	Response +QMTPUB: (range of supported <client_idx>s),(range of supported <msgID>s),(range of supported <qos>s),(list of supported <retain>s), <topic> ,(range of supported <msglen>s) OK
Write Command Publish variable-length messages AT+QMTPUB=<client_idx>,<msgID>,<qos>,<retain>,<topic>	Response > After > is reported, input the data to be sent. Press Ctrl+Z to send the data, or Esc to cancel the operation. OK +QMTPUB: <client_idx>,<msgID>,<result>[,<value>] If there is any error: ERROR
Write Command Publish fixed-length messages AT+QMTPUB=<client_idx>,<msgID>,<qos>,<retain>,<topic>,<msglen>	Response > After > is responded, input the data to be sent. The number of bytes of input data must equal <msglen> . OK +QMTPUB: <client_idx>,<msgID>,<result>[,<value>] If there is any error: ERROR
Maximum Response Time	<pkt_timeout> × <retry_times> (default 15 s), determined by the network

Characteristics

/

Parameter

<client_idx>	Integer type. MQTT client identifier. Range: 0–5.
<msgid>	Integer type. Message identifier of a packet. Range: 0–65535. It will be 0 only when <qos> =0.
<qos>	Integer type. QoS level at which the client wants to publish messages. 0 At most once 1 At least once 2 Exactly once
<retain>	Integer type. Determines whether the server will retain the message after it has been delivered to the current subscribers. 0 The server will not retain the message after it has been delivered to the current subscribers 1 The server will retain the message after it has been delivered to the current subscribers
<topic>	String type. Topic that needs to be published.
<msglen>	Integer type. Length of the message to be published. Range: 1–4096. Unit: byte.
<result>	Integer type. Command execution result. 0 Packet sent successfully and ACK received from the server (message that is published when <qos> =0 does not require ACK) 1 Packet retransmission 2 Failed to send a packet
<value>	If <result> is 1, it means the number of times a packet has been retransmitted. If <result> is 0 or 2, it will not be presented.
<pkt_timeout>	Integer type. Packet delivery timeout. Range: 1–60. Default value: 5. Unit: second.
<retry_times>	Integer type. Retry times when packet delivery times out. Range: 0–10. Default value: 3.

NOTE

1. If this command is executed successfully and **OK** is reported, the client can continue to publish new packets. The maximum quantity of the packet to be transmitted should not be greater than that of inflight windows (5).
2. After executing this command, the client will be ready to send data as a payload. The maximum length of data input at a time is 4096 bytes. To send the data, remember to press **Ctrl+Z**.
3. **PUBLISH** messages can be sent either from a publisher to the server, or from the server to a subscriber. When a server publishes messages to a subscriber, the following URC will be returned to notify the host to read the received data reported by the MQTT server:
+QMTRECV: <client_idx>,<msgid>,<topic>[,<payload_len>],<payload>
For more information on the URC description, see **Chapter 4.2**.

3.3.9. AT+QMTPUBEX Publish Messages (Command Mode)

This command publishes messages. It provides the same functions as **AT+QMTPUB**, except that the format is different.

AT+QMTPUBEX Publish Messages (Command Mode)	
Test Command AT+QMTPUBEX=?	Response +QMTPUBEX: (range of supported <client_idx>s),(range of supported <msgid>s),(range of supported <qos>s),(list of supported <retain>s),<topic>,<msg> OK
Write Command AT+QMTPUBEX=<client_idx>,<msgid>,<qos>,<retain>,<topic>,<msg>	Response OK +QMTPUB: <client_idx>,<msgid>,<result>[,<value>] If there is any error: ERROR
Maximum Response Time	<pkt_timeout> × <retry_times> (default 15 s), determined by the network
Characteristics	/

Parameter

<client_idx>	Integer type. MQTT client identifier. Range: 0–5.
<msgid>	Integer type. Message identifier of a packet. Range: 0–65535. It will be 0 only when <qos>=0.
<qos>	Integer type. QoS level at which the client wants to publish messages. 0 At most once 1 At least once 2 Exactly once
<retain>	Integer type. Determines whether the server will retain the message after it has been delivered to the current subscribers. 0 The server will not retain the message after it has been delivered to the current subscribers 1 The server will retain the message after it has been delivered to the current subscribers
<topic>	String type. Topic that needs to be published.
<msg>	String type. Message to be published. Maximum length: 560 bytes.
<result>	Integer type. Command execution result. 0 Packet sent successfully and ACK received from the server (message that is published when <qos>=0 does not require ACK) 1 Packet retransmission

	2 Failed to send a packet
<value>	If <result> is 1, it means the number of times a packet has been retransmitted. If <result> is 0 or 2, it will not be presented.
<pkt_timeout>	Integer type. Packet delivery timeout. Range: 1–60. Default value: 5. Unit: second.
<retry_times>	Integer type. Retry times when packet delivery times out. Range: 0–10. Default value: 3.

NOTE

The response flow of **AT+QMT PUBEX** will be the same as the response flow of **AT+QMT PUB**.

3.3.10. AT+QMTRECV Read Messages from Buffer

This command reads messages from the storage buffer where the messages are stored after being reported by the server.

AT+QMTRECV Read Messages from Buffer	
Test Command AT+QMTRECV=?	Response OK
Read Command AT+QMTRECV?	Response +QMTRECV: <client_idx>,<store_status_0>,<store_status_1>,<store_status_2>,<store_status_3>,<store_status_4> OK If there is no MQTT connection: OK
Write Command AT+QMTRECV=<client_idx>[,<recv_id>]	Response List of (+QMTRECV: <client_idx>,<msgid>,<topic>,[<payload_len>,<payload>])s OK If no message has been received: OK If there is no MQTT connection: ERROR
Maximum Response Time	Determined by the network.

Characteristics

/

Parameter

<client_idx>	Integer type. MQTT client identifier. Range: 0–5.
<store_status>	Integer type. Indicates if a message is stored in the buffer. As there can be maximum 5 messages in a buffer, the URC can report at most 5 messages simultaneously. 0 No stored messages 1 There are stored messages (up to 5)
<recv_id>	Integer type. Serial number of a received message. Range: 0–4. All buffer data will be read if this parameter is not specified.
<msgID>	Integer type. Message identifier of the packet. Range: 0–65535. It will be 0 only when <qos> =0.
<topic>	String type. Topic that needs to be published.
<payload_len>	Integer type. Payload length. Unit: Byte.
<payload>	String type. Payload relating to topic name.

4 MQTT URCs

This chapter describes the MQTT URCs.

4.1. +QMTSTAT: URC to Indicate State Change in MQTT Link Layer

The URC that begins with **+QMTSTAT** will be reported when the state of an MQTT link layer is changed.

+QMTSTAT: URC to Indicate State Change in MQTT Link Layer

+QMTSTAT: <client_idx>,<err_code>

When the state of an MQTT link layer is changed, the client will close the MQTT connection and report the URC.

Parameter

<client_idx>	Integer type. MQTT client identifier. Range: 0-5.
<err_code>	Integer type. Error code. Refer to the table below for more information.

Table 1: Error Codes of +QMTSTAT: URC

<err_code>	Description	Solution
1	Connection is closed or reset by a peer.	Execute AT+QMTOPEN command to reopen the MQTT connection.
2	Sending PINGREQ packet timed out or failed.	First, deactivate and activate the PDP. Then, reopen the MQTT connection.
3	Sending CONNECT packet timed out or failed.	1. Check whether the inputted username and password are correct. 2. Make sure the client ID is not used. 3. Reopen the MQTT connection and try to send CONNECT packet to the server again.
4	Receiving CONNACK packet timed out or failed.	1. Check whether the inputted username and password are correct. 2. Make sure the client ID is not used. 3. Reopen the MQTT connection and try to send CONNECT packet to the server again.

5	The client sends DISCONNECT packet to the sever and the server starts closing the MQTT connection.	This is a normal process.
6	The client takes the initiative to close the MQTT connection due to constant packet sending failure.	1. Make sure that the data are correct. 2. Try to reopen the MQTT connection since there may be a network congestion or an error.
7	The link is not alive, or the server is unavailable.	Make sure that the link is alive, or the server is currently available.
8–255	Reserved for future uses.	

NOTE

Error codes 5, 6 and 7 are currently not supported.

4.2. +QMTRECV: URC to Inform the Host to Read MQTT Packet Data

The URC that begins with **+QMTRECV:** is mainly used to inform the host to read the received MQTT packet reported from the MQTT server.

+QMTRECV: URC to Inform the Host to Read MQTT Packet Data

+QMTRECV: <client_idx>,<msgID>,<topic>[,<payload_len>],<payload>	Notify the host to read the received data reported from the MQTT server.
+QMTRECV: <client_idx>,<recv_id>[,<payload_len>]	Notify that the message received from the MQTT server has been stored in a buffer.

Parameter

<client_idx>	Integer type. MQTT client identifier. Range: 0–5.
<msgID>	Integer type. Message identifier of the packet.
<topic>	String type. Topic that has been received from the MQTT server.
<payload_len>	Integer type. Payload length. Unit: byte.
<payload>	String type. Payload relating to topic name.
<recv_id>	Integer type. Serial number of a received message. Range: 0–4.

5 Examples

This chapter provides some examples that explain how to use the MQTT AT commands.

5.1. Example of MQTT Operation without SSL

```

AT+QMTCFG="aliauth",0,"oyjtmPI5a5j","MQTT_TEST","wN9Y6pZSIly7Exa5qVzcmigEGO4kAazZ"
//Configure Alibaba device information for AliCloud.

OK
AT+QMTOPEN=?
+QMTOPEN: (0-5),<host_name>,(0-65535)

OK
AT+QMTOPEN=0,"iot-as-mqtt.cn-shanghai.aliyuncs.com",1883 //Open a network for MQTT client.
OK

+QMTOPEN: 0,0 //The MQTT client network opened successfully.
AT+QMTOPEN?
+QMTOPEN: 0,"iot-as-mqtt.cn-shanghai.aliyuncs.com",1883

OK
AT+QMTCONN=?
+QMTCONN: (0-5),<clientId>,<username>,<password>

OK

//Connect a client to the MQTT server.
//If AliCloud is connected, AT+QMTCFG="aliauth" can be used to configure the device information in
advance. There is no need to provide the username/password from now on.
AT+QMTCONN=0,"clientExample"
OK

+QMTCONN: 0,0,0 //The client has successfully connected to the MQTT server.
AT+QMTSUB=?
+QMTSUB: (0-5),(1-65535),<topic>,(0-2)

```

OK

//Subscribe to topics.

AT+QMTSUB=0,1,"topic/example",2

OK

+QMTSUB: 0,1,0,2

AT+QMTSUB=0,1,"topic/pub",0

OK

+QMTSUB: 0,1,0,0

//If a client subscribed to a topic and other devices publish the same topic to the server, the module will report the following information.

+QMTRECV: 0,0,"topic/example","This is the payload related to topic"

//Unsubscribe from topics.

AT+QMTUNS=0,1,"topic/example"

OK

+QMTUNS: 0,2,0

AT+QMTPUB=?

+QMTPUB : (0-5),(0-65535),(0-2),(0,1),<topic>,(1-4096)

OK

//Publish messages.

AT+QMTPUB=0,0,0,0,"topic/pub"

>These are test data, hello MQTT. //After receiving **>**, input data **These are test data, hello MQTT.** and then send it. The maximum length of the data to be sent is 4096 bytes. All the data beyond 4096 bytes will be omitted. After inputting data, press **Ctrl+Z** to send them.

OK

+QMTPUB: 0,0,0

//If a client subscribed to a topic named "topic/pub" and other devices publish the same topic to the server, the module will report the following information.

+QMTRECV: 0,0,"topic/pub","These are test data, hello MQTT."

AT+QMTDISC=0 //Disconnect a client from the MQTT server.

OK

+QMTDISC: 0,0 //The connection closed successfully.

5.2. Example of MQTT Operation with SSL

//Configure the MQTT session into SSL mode.

AT+QMTCFG="ssl",0,1,2

OK

//If SSL authentication mode is intended to be set as "manage server and client authentication if requested by the remote server" (<seclvl>=2 in **AT+QSSLCFG**), upload a server root CA certificate, a client certificate, and a client private key to UFS.

AT+QFUPL="cacert.pem",1758,100 //Upload a CA certificate to UFS.

CONNECT

<Input the cacert.pem data, and the size is 1758 bytes>

+QFUPL: 1758,384a

OK

AT+QFUPL="client.pem",1220,100 //Upload a client certificate to UFS.

CONNECT

<Input the client.pem data, and the size is 1220 bytes>

+QFUPL: 1220,2d53

OK

AT+QFUPL="user_key.pem",1679,100 //Upload a client private key to UFS.

CONNECT

<Input the user_key.pem data, and the size is 1679 bytes>

+QFUPL: 1679,335f

OK

//Configure the path of the CA certificate for SSL context 2.

AT+QSSLCFG="cacert",2,"cacert.pem"

OK

//Configure the path of the client certificate for SSL context 2.

AT+QSSLCFG="clientcert",2,"client.pem"

OK

//Configure the path of the client private key for SSL context 2.

AT+QSSLCFG="clientkey",2,"user_key.pem"

OK

//Configure the authentication mode for SSL context 2.

AT+QSSLCFG="seclvl",2,2

//SSL authentication mode: server and client authentication if requested by the remote server.

```

OK
AT+QSSLCFG="sslversion",2,4           //SSL authentication version
OK
AT+QSSLCFG="ciphersuite",2,0XFFFF     //Cipher suite
OK
AT+QSSLCFG="ignorelocaltime",2,1      //Ignore the authentication time.
OK

//Start the MQTT SSL connection.
AT+QMTOPEN=0,"a1zgnxur10j8ux.iot.us-east-1.amazonaws.com",8883
OK

+QMTOPEN: 0,0

//Connect to the MQTT server.
AT+QMTCONN=0,"MQTT-1"
OK

+QMTCONN: 0,0,0

//Subscribe to topics.
AT+QMTSUB=0,1,"$aws/things/ MQTT-1/shadow/update/accepted",1
OK

+QMTSUB: 0,1,0,1

//Publish messages.
AT+QMTPUB=0,1,1,0,"$aws/things/MQTT-1/shadow/update/accepted"
>These are published data from client
OK

+QMTPUB: 0,1,0

//If a client subscribed to a topic named "$aws/things/MQTT-1/shadow/update/accepted" and other devices
publish the same topic to the server, the module will report the following information.
+QMTRECV: 0,1,"$aws/things/MQTT-1/shadow/update/accepted","These are published data from
client"

//Disconnect a client from the MQTT server.
AT+QMTDISC=0
OK

+QMTDISC: 0,0

```

6 Appendix References

Table 2: Related Documents

Document Name
[1] MQTT V3.1 Protocol Specification
[2] MQTT V3.1.1 Protocol Specification
[3] Quectel_BG770A-GL&BG95xA-GL_SSL_Application_Note

Table 3: Terms and Abbreviations

Abbreviation	Description
ACK	Acknowledgement
CA	Certificate Authority
IP	Internet Protocol
MQTT	Message Queuing Telemetry Transport
PDP	Packet Data Protocol
QoS	Quality of Service
SSL	Secure Sockets Layer
TCP	Transmission Control Protocol
UFS	User File System
URC	Unsolicited Result Code